

Study Buddy AI — Comprehensive Project Documentation

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1. Executive Overview

Study Buddy AI is a full-stack, AI-driven learning web application designed to help students transform lecture notes, textbook chapters, and documentation files into structured, interactive study materials.

By uploading documents in **PDF**, **TXT**, or **MD** format, students automatically receive:

- **Executive Summaries & Deep-Dive Concept Breakdowns**
- **Interactive 3D Active Recall Flashcards**
- **Self-Grading Practice Quizzes with Explanations & "Try Again" Mode**
- **Customized Multi-Day Study Plans with Daily Workload Targets**
- **Gamified Study Streaks & Learning Analytics**

2. Tech Stack & Architecture

Frontend (Client)

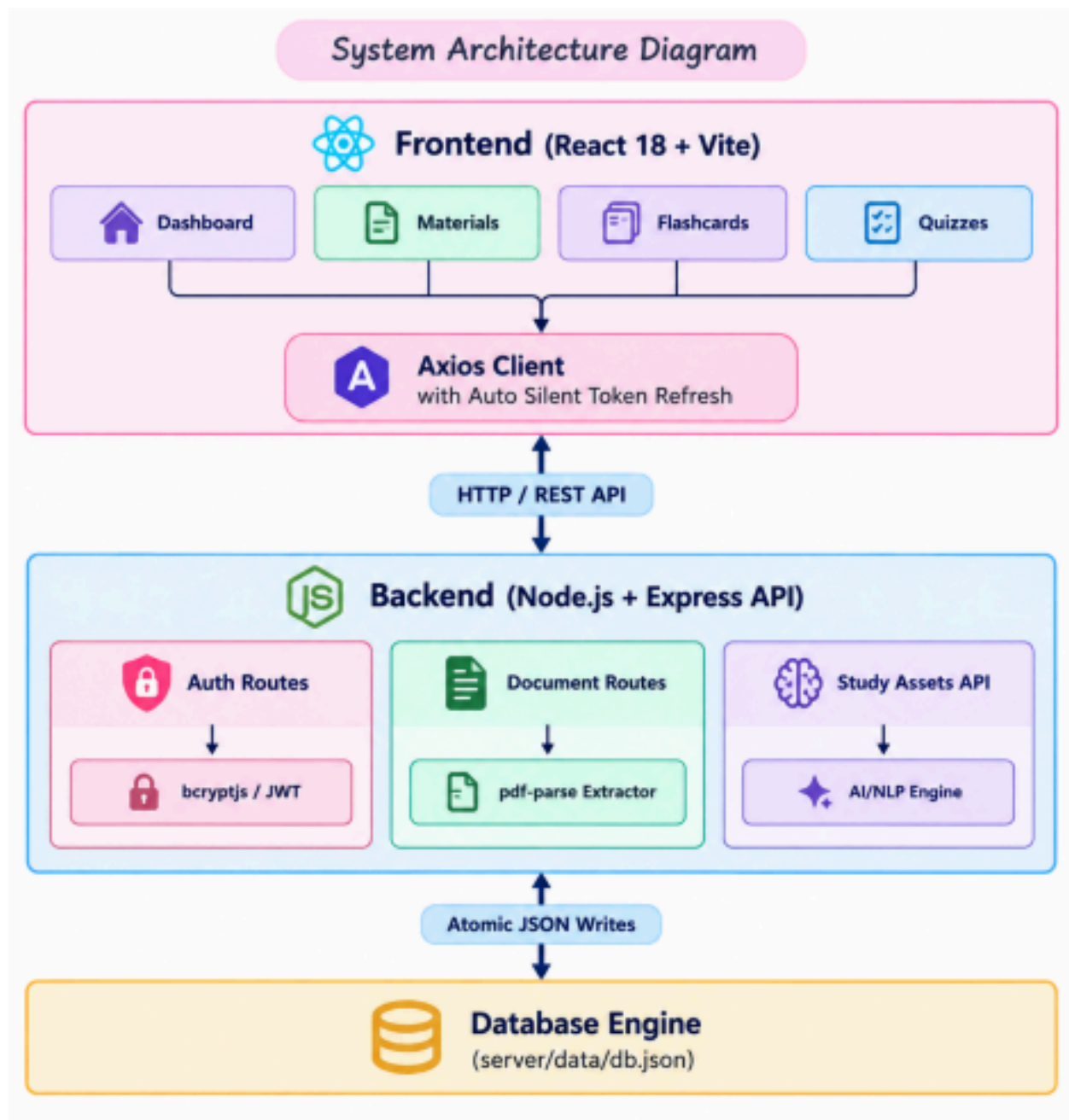
- **Framework:** React 18 with Vite
- **Routing:** React Router DOM v6

- **Styling:** Tailwind CSS v3, CSS 3D Transforms, Glassmorphism design system -
- State Management:** React Context API ([AuthContext](#), [ToastContext](#)) - **HTTP Client:** Axios with automatic 401 Silent Token Refresh Interceptors - **Icons & Visual Effects:** Lucide React, Canvas Confetti

Backend (Server)

- **Runtime:** Node.js (v24+)
- **Web Framework:** Express.js
- **Authentication:** JWT (Short-lived Access Tokens + HTTP-only Refresh Tokens) with bcryptjs password hashing
- **File Processing:** Multer for disk storage, [pdf-parse](#) for PDF text extraction -
- Database Engine:** Custom JSON File Storage with atomic writes ([server/data/db.json](#))
- **AI Heuristic & NLP Service:** Contextual text parsing engine supporting fallback NLP and OpenAI/Gemini API integrations.

3. System Architecture Diagram



4. Core Features & Functional Specifications

❖❖ Authentication & Session Refresh

- **User Registration:** Captures name, email, password, primary study goal, and daily target hours. Passwords are salted and hashed using **bcryptjs**.
- **JWT Dual Token Security:**
 - **Access Token:** Short-lived (15 mins), passed in **Authorization: Bearer <token>** header.
 - **Refresh Token:** Long-lived (7 days), stored securely in HTTP-only cookies and recorded in DB for token rotation.
- **Silent Token Refresh Interceptor:** When an API call returns **401 Unauthorized** with **tokenExpired: true**, the frontend Axios interceptor automatically requests a new access token in the background and retries the original failed request seamlessly without

interrupting the user.

?? File Upload & Text Extraction

- Supports Drag & Drop and manual browsing for **.pdf**, **.txt**, and **.md** files up to **15 MB**. - Validates mime types and extensions on both client and server.
- Parses PDF pages into clean, normalized text blocks using **pdf-parse**, calculating total word count and estimated reading times.

?? AI Summary Generation

Generates multi-level study breakdowns for every document:

1. **Executive Overview**: High-level abstract of the subject matter.
2. **Key Concepts & Takeaways**: Bulleted core principles.
3. **Key Terminology**: Essential domain terms with definitions.
4. **Deep-Dive Sections**: Structured section-by-section breakdown.

?? Interactive 3D Flashcards

- **3D Card Flip Animation**: Smooth CSS 3D perspective flip showing Question on front and Answer + Hint on back.
- **Active Recall Rating**: Users can mark cards as **Mastered** (Emerald) or **Needs Review** (Amber).
- **Deck Control**: Next/Previous navigation, card shuffle, and overall progress percentage.

?? Practice Quiz Engine & Scoring

- **Automated Question Generation**: Extracts core statements and creates fill-in-the-blank or multiple-choice questions with plausible distractors.
- **Interactive Exam Mode**: Question-by-question selection with real-time answer locking upon submission.
- **Instant Grading & Feedback**: Calculates percentage score %, displays pass/fail status (passing score = 70%), fires celebratory confetti on high scores, and provides detailed explanations for each question.
- **"Try Again" Functionality**: Retake quizzes, reset choices, and track score history over time.

?? Dynamic Study Plan Generator

- Converts document length and user-selected target days into a daily schedule. - Allocates specific daily focus topics, estimated completion times, and checkable task items.

?? Profile & Study Streak Metrics

- Tracks continuous daily study streaks.
- Displays learning statistics (Total Documents, Mastered Cards Count, Average Quiz Score %).
- Allows updating user goals, target study hours per day, and password changes.

Material Cleanup & Deletion

- Complete file manager for uploaded materials.
- Safe deletion removes the document entry, physically deletes the file from the server disk (`server/uploads/`), and purges associated summaries, flashcards, quizzes, and study plans.

✗ Toast Notification & Error System

- Centralized slide-in notification toast system (`ToastContext.jsx`).
- Categorized alerts: Success (Green), Error (Red), Warning (Amber), Info (Blue).

5. Database Schema & Data Models

Stored in persistent JSON file structure at `server/data/db.json`:

`Users`

```

"users": [
  {
    "id": "muf241r9lwo79",
    "createdAt": "2026-09-24T04:53:40.965Z",
    "updatedAt": "2026-09-24T08:08:29.306Z",
    "name": "sandhiya",
    "email": "p.sandhiya0404@gmail.com",
    "password": "$2a$10$b3N7VwkzBdI.h9RdYVROy..WnXbsxIZHeeMU203x/hlbc",
    "avatar": "https://api.dicebear.com/7.x/bottts/svg?seed=sandhiya",
    "studyGoals": "Ace exams and build solid understanding",
    "dailyTargetHours": 2,
    "studyStreak": 1,
    "lastLoginDate": "2026-09-24T04:53:40.965Z"
  },
  {
    "id": "muf9cs05fjdyz",
    "createdAt": "2026-09-24T08:16:25.541Z",
    "updatedAt": "2026-09-24T08:16:25.541Z",
    "name": "kalai",
    "email": "kalai@gmail.com",
    "password": "$2a$10$gKMwxpiw8.kzoru3WiyN5.uoPur0/WNXCYr7cl09MsiJ8",
    "avatar": "https://api.dicebear.com/7.x/bottts/svg?seed=kalai",
    "studyGoals": "Ace exams and build solid understanding",
    "dailyTargetHours": 2,
    "studyStreak": 1,
    "lastLoginDate": "2026-09-24T08:16:25.541Z"
  },
  {
    "id": "mufu2bdy3ix2o",
    "createdAt": "2026-09-24T17:56:09.382Z",
    "updatedAt": "2026-09-24T17:56:09.382Z",
    "name": "BOTVA"
  }
]

```

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Documents

```

    },
    "documents": [
      {
        "id": "muf356f4079qu",
        "createdAt": "2026-09-24T05:22:33.280Z",
        "updatedAt": "2026-09-24T05:22:33.290Z",
        "userId": "muf241r9lwo79",
        "originalName": "Routing Algorithm in Computer Networks.pdf",
        "filename": "Routing_Algorithm_in_Computer_Networks-1790227352907",
        "filePath": "C:\\Users\\acer\\Downloads\\Study Buddy 2\\server\\upl",
        "fileType": ".pdf",
        "fileSize": 396081,
        "extractedText": "Home Data_communication_computer_network Routing",
        "wordCount": 363,
        "pageCount": 2,
        "summary": {
          "title": "Study Overview: Routing Algorithm in Computer Networks",
          "wordCount": 363,
          "readingTimeMinutes": 2,
          "executiveSummary": "Home Data_communication_computer_network",
          "keyPoints": [
            "Home Data_communication_computer_network Routing Algorithm",
            "Routing algorithm mathematically computes the best path, i.e.",
            "It constructs the routing\\ntable depending upon the network t",
            "Isolated algorithm This algorithm procures the routing inform",
            "The\\nstatic routing table is constructed based upon the routi"
          ],
          "keyTerminology": [
            {
              "term": "Routing",

```

Ln 4, Col 5 Spaces: 2 UTF

quizzes

```

    ],
    "quizzes": [
      {
        "id": "muf356f9sy2cd",
        "createdAt": "2026-09-24T05:22:33.285Z",
        "updatedAt": "2026-09-24T07:18:31.924Z",
        "userId": "muf241r9lwo79",
        "docId": "muf356f4079qu",
        "docName": "Routing Algorithm in Computer Networks.pdf",
        "title": "Quiz: Routing Algorithm in Computer Networks.pdf",
        "questions": [
          {
            "id": "q_3",
            "questionText": "Fill in the blank: \"After a data packet leaves a node, it is either dropped or it is forwarded to the next node. The decision is based on the state of the node and the state of the network. The state of the node is determined by the state of the node's resources and the state of the node's neighbors. The state of the network is determined by the state of the network's resources and the state of the network's topology. The state of the node's resources is determined by the state of the node's CPU, memory, and network interface. The state of the node's neighbors is determined by the state of the neighbors' CPU, memory, and network interface. The state of the network's resources is determined by the state of the network's CPU, memory, and network interface. The state of the network's topology is determined by the state of the network's links and the state of the network's nodes.",
            "options": [
              "choose",
              "Static dependency",
              "Variable threshold",
              "Sequential outcome"
            ],
            "correctAnswerIndex": 0,
            "explanation": "The correct term is \"choose\", which completes the sentence."
          },
          {
            "id": "q_4",
            "questionText": "Fill in the blank: \"Routing algorithm mathematical models are used to determine the best path for a data packet to travel from a source node to a destination node. The mathematical models are based on the state of the network and the state of the network's resources. The state of the network is determined by the state of the network's resources and the state of the network's topology. The state of the network's resources is determined by the state of the network's CPU, memory, and network interface. The state of the network's topology is determined by the state of the network's links and the state of the network's nodes.",
            "options": [
              "Static dependency",
              "Sequential outcome",
              "External factor",
              "Variable threshold"
            ],
            "correctAnswerIndex": 2,
            "explanation": "The correct term is \"Sequential outcome\", which completes the sentence."
          }
        ]
      }
    ]
  }
}

```

6. API Endpoints Reference

Method	Endpoint	Description	Auth Required
POST	/api/auth/register	Register new user account	✗

Method	Endpoint	Description	Auth Required
POST	/api/auth/login	Authenticate user & issue tokens	✗

POST	/api/auth/refresh-token	Issue new access token using refresh token	✗
POST	/api/auth/logout	Revoke session & clear cookies	✗
GET	/api/auth/me	Fetch active user profile & analytics	✓
PUT	/api/auth/profile	Update profile settings / password	✓
POST	/api/documents/upload	Upload PDF/TXT/MD & extract assets	✓
GET	/api/documents	List all user uploaded documents	✓
GET	/api/documents/:id	Get document details & extracted text	✓
DELETE	/api/documents/:id	Delete document & associated assets	✓
GET	/api/flashcards/:docId	Fetch flashcards for document	✓
PATCH	/api/flashcards/:setId/card/:cardId	Update card mastery status	✓
GET	/api/quizzes/:docId	Fetch quiz questions & attempt history	✓
POST	/api/quizzes/:quizId/submit	Submit quiz answers & return score	✓
GET	/api/study-plans/:docId	Fetch structured study schedule	✓

Method	Endpoint	Description	Auth Required
PATCH	/api/study-plans/:planId/task/:taskId	Toggle task completion checkbox	✓

7. Directory Structure



8. Setup & Execution Guide (VS Code)

Prerequisites

- **Node.js:** v18.0.0 or higher (v24 recommended)
- **npm:** v9.0.0 or higher

Quick Start Instructions

1. **Open Workspace:** Open `c:\Users\acer\Downloads\Study Buddy 2` in VS Code.
2. **Install Dependencies:** Open a terminal in VS Code and run:

```
npm run install-all
```

3. **Run Concurrent Development Server:**

```
npm run dev
```

4. **Access App:** Open browser to <http://localhost:5173/>

```
PS C:\Users\acer\Downloads\Study Buddy 2> cd "C:\Users\acer\Downloads\Study Buddy 2"
PS C:\Users\acer\Downloads\Study Buddy 2> cd client
PS C:\Users\acer\Downloads\Study Buddy 2\client> npm run dev

> study-buddy-client@1.0.0 dev
> vite

Port 5173 is in use, trying another one...

VITE v5.4.21 ready in 596 ms
  → Local:   http://localhost:5174/
  → Network: use --host to expose
  → press h + enter to show help
```

9.Window PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> cd "$HOME\Downloads\Study Buddy 2\client"
PS C:\Users\acer\Downloads\Study Buddy 2\client> npm run dev

> study-buddy-client@1.0.0 dev
> vite

VITE v5.4.21 ready in 620 ms
  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help
```

API TESTING LINK: 📺 Recording 2026-09-25 125543.mp4

Demo Video Link: 📺 Video Project (1) demo.mp4

Code Drive Link: 📁 VCR

GithubLink:<https://github.com/canrani/TN-AI-AUGMENTED-BACKEND-DEVELOPMENT.git>